

Monthly outcome: a manipulator under human supervision

M17 • 15 Feb 2028 - 19 Mar 2028 • 75 hours

Build a small, reproducible testbed for grasping and transferring an object. Treat it as an integrated robotic system connecting kinematics, planning, operator actions, data, commands, updates and engineering responsibility. The M17 deliverables are a simulated transfer of a simple object, a study of repeatability and failures, an interface for selecting modes and cancelling actions, and a documented assessment of threats and hazards.

Four areas of work

This month covers manipulation, human-robot interaction, ethics and cybersecurity, industrial applications and teamwork. Allocate the manipulator work so that every topic receives attention. A complex anthropomorphic hand, tactile skin and force-controlled grasping are not required here. Under the curriculum, the advanced OPT block on hand mechanics and tactile sensors takes 120 hours after M17 and is separate from these 75 hours.

What to bring from M14-M16

From M14, use the object-observation format and the descriptions of coordinate frames and uncertainty; from M16, use the prepared manipulator scene, collision checks, pre-grasp pose and trajectory validation. In the main experiment, specify the object's position precisely in simulation to distinguish manipulation errors from vision errors. Then deliberately offset the estimated position and orientation. A complete vision-guided grasping system is outside this month's required project.

Testbed and working environment

Prepare an existing manipulator model with a simple parallel gripper, a table, a cube or cylinder, and a target region. Work in Ubuntu 24.04 with ROS 2 Jazzy and a compatible MoveIt 2 configuration. You may additionally test your own mechanism once the baseline simulation experiment works; rated limits and real contacts require separate tests. A cancel button in a graphical interface does not, by itself, constitute a hardware emergency stop.

By the end of the month, explain why the manipulator might fail to reach its goal, who is authorised to issue commands and how the operator knows the current mode. State which data are retained, what happens when communication is lost and what must be checked before changing a limit. For the future hexapod, the same questions concern body motion, leg contacts and handing control to a human.

75 hours and a separate hand project

Four blocks

1-20: FK and IK, grasp coordinate frames, object transfer, contact and repeatability; 10 hours of theory and analysis +10 of practice. 21-40: teleoperation, levels of autonomy, modes, operator feedback and a small interface study; 10+10. 41-60: privacy, data errors, threat modelling, authentication, encryption and updates; 10+10. 61-75: PLCs, industrial buses, standards, requirements, verification and planning a team exercise; 5+10. Total: 35 hours of analysis and 40 of practice.

In the first three blocks, schedule reading in 2-hour sessions; in the last block, use 2+2+1. Divide practical work into four Sunday blocks of 10 hours, or equal periods that suit your schedule. Page dates follow the original curriculum, while step numbers indicate the learning workload. Overlapping module boundaries, rest, holidays and separate checkpoints do not add hours; OPT120 is counted separately.

Keeping the scope manageable

Start with one robot, one object type and one scene, then introduce variations. Keep development of custom motor control, vision and a physical hand outside this experiment. MoveIt plans the geometric sequence of actions; assess contact and object retention using separate simulator observations. A simplified object attachment demonstrates planning, but does not establish that grasping force is sufficient.

Preparation in the first block

Create a model-description file: simulator and MoveIt versions, object dimensions, joint limits, table position, coordinate frames and the source of the pose estimate. For each experiment, save `run_id`, the random-number generator seed, parameters, initial object state and result. Provide observation without issuing commands, manual control, plan execution and stopping. Maintain a shared log of `command_id`, mode and actor, even while there is only one operator.

Working folders: manipulation/ for tasks and scenes; operator/ for the interface and study protocol; evaluation/ for repeated experiments; review/ for hazards, threats and ethical decisions; requirements/ for criteria and changes. Store secret keys and participants' personal data outside the repository. In the report, distinguish observations, proposed causes and ways to test them: this helps another engineering team understand the system.

Manipulation: poses, approach and contact

What to study

Review forward kinematics, FK: it determines end-effector position and orientation from joint coordinates. Inverse kinematics, IK, solves the reverse problem; there may be several valid solutions or none. Define the world, base, tool, object and grasp coordinate frames. Define the grasp pose relative to the object and the pre-grasp pose as an offset along the approach direction. Before connecting the planner, verify transform composition and inversion using three known poses.

Transfer sequence: move into position → approach → close the gripper → confirm retention → lift → transfer → lower → open → withdraw. Define geometric and logical conditions for each stage. The attach operation in the planning scene changes the collision model, but does not measure contact. When introduced to force and impedance control, establish how position control differs from specifying a desired interaction; a complete force-control loop is studied separately.

Task 1. FK, IK and the grasp coordinate frame

Use an existing manipulator model and three reachable object poses. Specify the grasp offset, calculate the tool pose and solve IK while checking joint limits and self-collisions. Calculate FK for the solution and compare its position and orientation with the target. Save a table of poses, solutions and rejection reasons. Test an object outside the workspace and a rotation of 180°; if no valid IK solution exists, the limits must remain enabled.

Task 2. A complete grasp-and-transfer cycle

Use MoveIt Task Constructor or a compatible sequence of ROS actions. Save each stage and its result, add the table and object to the scene, and check attached geometry during transfer. Prepare a task description, configuration and reproducible demonstration. Separately verify that the object reached the target region and was released, and that the arm withdrew: successfully producing a plan without executing it must not count as task success.

Interpreting contact checks

For a physical simulation of retention, specify mass, friction and the gripper model; check whether the object can fall. When using a simplified attach, label the mode a kinematic demonstration (kinematic demo). Consider a virtual spring $F=K \cdot (x_{\text{des}} - x)$, with K in N/m, and a damper. Explain why high stiffness and delay can increase contact forces. Specified forces and speeds are educational parameters and do not replace checking rated specifications and conducting physical tests.

Repeatability and operator interaction

Task 3. Defining a successful transfer

Perform 10 runs with the same initial scene and 10 with small, predefined object offsets. Save success status and failure stage, final pose, cycle time and retention state. Before the experiments, specify the target region in metres and acceptable orientation. Present every result, mean error and dispersion; if deterministic simulation produces identical results, explain why.

Repeatability describes how close results are under identical conditions; accuracy describes closeness to a specified reference. An incorrect grasp offset can make the arm place the object in the same wrong location every time: test this case. In separate experiments, introduce an incorrect mass, friction or calibration offset and distinguish these causes. In a kinematic demonstration, evaluate the plan and pose; such an experiment does not measure holding force.

Human-robot interaction (HRI)

Study direct teleoperation, assisted control, autonomous task execution and explicit transfer of control. The operator needs the current mode, active goal, age of state data, reason for rejection and available action. A mode error occurs when a person expects manual control while an autonomous plan is active, or the reverse. Spatial interaction requires clearly indicated access zones and the robot's future path; low speed alone does not demonstrate contact safety.

Task 4. Modes and operator intervention

Create a simple panel or CLI with view, manual, auto and stopped states. Include `command_id` and mode in commands. Before a manual↔auto transition, complete or cancel the previous ROS action; after stopped, the old queue must not resume automatically. Save the transition table, interface and test log. Test a manual request during auto and a repeated start after communication loss.

Give the operator a scenario: position the gripper, start a transfer and stop it when the object is wrong. Display the planned path and current stage; present debugging diagnostics separately. The same principle applies to transferring control of gait: a human action must bring the system into a clear state, otherwise competing commands may enter a shared topic.

Interface testing and privacy

Task 5. Test the interface on a clearly defined task

Conduct a pilot study in simulation with two interface versions: for example, buttons alone and buttons with explicit mode and state indications. Use the same task: start a transfer, notice an intentionally incorrect goal and cancel the action. Explain the task and the data to be recorded to the volunteer beforehand. Conduct this experiment without physical robot motion near a person. If no participant is available, follow a predefined scenario yourself and state that no independent study took place.

Prepare 3-5 attempts for each version, alternate their order and provide the same initial training. Record the time until correct cancellation, mode errors, unnecessary actions and a brief answer to “what is the robot doing now?”. Save the protocol, anonymised table and three observations. So few attempts cannot establish statistical superiority; use them to correct specific unclear indications.

Task 6. Camera data and model errors

List the testbed's data: table images, object pose, operator actions, identifiers and logs. For each type, state its purpose and retention period, access rights and deletion method. In the demonstration, retain only the necessary table region or use synthetic data. If people or personal belongings appear in the frame, exclude unnecessary regions from the educational dataset in advance; transfer to an external service requires an appropriate basis.

Return to the baseline CNN from M14 and compare errors under two lighting conditions and with two object materials. Consider practical ethical questions: for whom and under which conditions does the system perform worse, who knows its limitations and who can correct the result? If the data contain no people, evaluate gaps in operating conditions and interface accessibility; there is no basis here for demographic statistics. Good average performance does not establish performance under unstudied conditions.

A documented decision

Write a short record: “question → alternatives → chosen decision → verification → remaining limitation”. For example, events and anonymised errors suffice to measure cancellation time, so operator video need not be retained. If pose-estimation confidence is low, request another observation before grasping. Include these decisions and supporting results in the final engineering review.

Command security: a threat model for your testbed

Distinguish hazards from threats

A hazard is a potential source of harm, such as unexpected gripper motion. A cybersecurity threat is the possibility of malicious interference, such as an unauthorised command or a substituted update. A code error, malfunction or attack may lead to the same consequence. For each case, record the cause, protection and test. Encryption does not correct an invalid joint limit, and a limit switch does not determine access rights to video recordings.

Task 7. A threat model for your own testbed

Draw the testbed boundaries: operator, ROS and DDS network, control process, storage and update channel. For each flow, identify the protected asset, sender, recipient, trust boundary and possible violation. Analyse unauthorised commands, replay of old commands, calibration changes, corrupted updates and log leaks. Prepare a table of 5-8 specific threats with priorities and owners of protective measures.

In an isolated experiment, test commands containing an unknown actor, expired timestamp, repeated command_id, NaN and out-of-range values. The expected result is rejection with an explanation and no simulated motion. DDS certificates, identity and permissions govern admission to the system and its channels; the application checks command freshness and prevents repeated execution. Enabling SROS2 alone does not protect against all these threats.

Task 8. Minimum protection and an update

Study authentication, authorisation, encryption and trust in updates. On the local testbed, configure minimum permissions for one publisher and subscriber using SROS2, or reproduce the official example in a separate namespace. Test permitted and prohibited publication. Save a public description of roles and permissions and verification logs; exclude private keys from the final package.

Prepare an educational configuration-update scenario. A new version of the limits requires a contents description, a hash and engineering review; a deliberately modified file must be rejected, and the previous working version must be restored in simulation. Explain that a hash detects change, but verifying authorship requires a trusted signature or channel. Parameters may change during motion only through a defined state transition.

Industrial applications: the cell and its requirements

What to understand at an introductory level

A PLC executes industrial cycle logic and processes inputs and outputs; the robot controller controls drives and the programme; a supervisory computer may plan actions and exchange data. An industrial bus connects devices and defines communication properties. Learn the purposes of EtherCAT, PROFINET and CAN; implementing these protocols is outside this month's scope. Distinguish ordinary control from separate protective functions.

ISO 10218 concerns industrial robots and their applications; ISO 13849 addresses safety-related parts of control systems; IEC 61508 provides the context of functional safety and its lifecycle. In M17, read official descriptions of scope and terminology, without claiming compliance with a standard. Applicability, the required level and verification of a real cell require separate input data and competent assessment.

Task 9. Sketch an educational cell

Draw the robot, table, object feed, presence sensor, operator panel and a notional PLC. Define a ready/request/busy/done/fault state exchange with cycle_id and test it using software process simulators. Save the diagram, I/O table and event sequence. Test a repeated request, a missing part, communication loss and reset during busy. The done state belongs to a particular cycle_id: an old event must not confirm processing of a new part.

Task 10. Requirements, changes and teamwork

Formulate 8-10 requirements: start conditions, acceptable scene, cancel behaviour, pose-estimate age, operator rights, logging and rejection on recovery. For each, provide a criterion and a link to its test. Change the grasp offset or speed and prepare a change request: reason, affected requirements, checks and restoration of the previous version. Support engineering review with data; label your own verification as self-review.

Plan how to gain teamwork experience: which part of the work can be submitted for review, which questions to ask a control or mechanical specialist, and which small contribution to a robotics project would demonstrate collaboration. Prepare a review package and a list of missing experience, such as industrial equipment, initial hardware commissioning and working with someone else's requirements. An educational testbed does not replace an internship or a verified professional qualification.

Eight checks for manipulation, the operator and threats

Manipulator and interface

S1 - unreachable pose: IK or the planner reports failure, and the system does not enter executing. S2 - the object touches the table during transfer: the collision model includes the attached object and rejects the plan. S3 - incorrect grasp offset: the robot misses repeatedly, and the report distinguishes repeatability from accuracy. In a physical simulation, add slipping; in a kinematic demonstration, attach does not confirm retention.

S4 - mode change during auto: a manual request triggers cancel, waits for coordinated completion and only then allows a manual command. Mode outputs are not mixed. S5 - loss of communication or pose estimates: data age increases, and the task receives the defined stopped or uncertain status. Network recovery does not start the old queue. Describe the simulator's actual response; this experiment does not demonstrate a certified protective stop.

Commands, data and the cycle

S6 - repeated command or unknown actor: the application rejects the request without repeating the cycle. Check `command_id`, mode and validity period separately from DDS participant identification. S7 - configuration change: integrity checking detects substitution, the update is not activated and the previous version can be reproduced. Separately state that without a trusted signature, forged authorship remains possible.

S8 - late done from the previous PLC cycle: the new request has a different `cycle_id`, so the old event does not count as success. Add an absent object after a claimed grasp: the hold-check stage must report the corresponding failure. Inspect the final archive manually: it must contain only necessary frames and events, without participants' personal data or secret keys.

Recording criteria

For each scenario, record the initial scene, triggering event, expected states, observed response, timing and remaining risk. Justify pose and timing tolerances as parameters of the educational task. This table does not define safe speeds for physical interaction with a person. The absence of errors in 10 repetitions does not prove zero failure probability.

Indicate how each result is supported: simulation, logic or document review, or a required physical test. Show the operator the reason for failure, whether motion has been cancelled and whether restarting is possible. If the execution outcome is still unknown, state that explicitly: hidden uncertainty prevents an accurate assessment of robot state.

Sessions 1-40: object transfer and the operator

Hours 1-10: five sessions of 2 hours

1-2: review FK and IK, coordinate frames and grasp offset; calculate three transforms by hand. 3-4: pre-grasp pose, approach, attach/detach and scene geometry. 5-6: transfer stages and criteria for planning and execution. 7-8: contact, friction, fundamentals of force and impedance control, and differences between kinematic and physical demonstrations. 9-10: repeatability, accuracy, test protocol and simulation limitations. Prepare the scene for the practical session.

Hours 11-20: practical session 1

11-12: Task 1, reachable and unreachable poses with FK verification. 13-14: Task 2, initial stages using Task Constructor or ROS actions. 15-16: complete transfer with the correct scene and target-region checks. 17-18: Task 3, repeated trials and pose variations. 19-20: failures S1-S3, grasp offset and cause analysis. If the physical grasp model is not ready, save the basic kinematic demonstration and state its limitations explicitly; the remaining weeks are needed for other topics.

Hours 21-30: five sessions of 2 hours

21-22: teleoperation, automated assistance, autonomy and command authority. 23-24: mode FSM, operator intervention, cancel and transition confirmation. 25-26: operator feedback, data age and mode errors. 27-28: spatial interaction, zones and limitations of the educational simulation. 29-30: pilot study, metrics, participant preparation and limitations of a small sample.

Hours 31-40: practical session 2

31-32: Task 4, a simple interface and mode table. 33-34: operator intervention during auto, communication loss and command_id log. 35-36: Task 5, two interface versions and pilot attempts. 37-38: analyse events, cancellation time and mode errors; fix one identified problem. 39-40: repeat the scenario and write a short HRI report stating its limitations.

If no volunteer is available, follow a previously written scenario yourself and list questions for external testing. Study results must reflect the attempts actually conducted. A simple version can use a CLI with prominent mode and state indications: this experiment focuses on making robot control understandable.

Sessions 41-75: data, threats and project handover

Hours 41-50: theory in 2-hour sessions

41-42: necessary data, privacy and retention periods. 43-44: machine-learning limitations, incomplete data and responsibility for decisions; connection to M14. 45-46: hazards, cybersecurity threats, protected assets and trust boundaries. 47-48: authentication, authorisation, encryption and command-freshness checks. 49-50: update integrity, signatures, restoration of the previous version and documentation of remaining risks.

Hours 51-60: practical session 3

51-52: Task 6, data inventory and checks across lighting conditions and materials. 53-54: Task 7, a flow diagram and five specific threats. 55-56: test commands with repetitions, NaN and an unknown actor; rejection logs. 57-58: Task 8, permitted and prohibited communication in an isolated SROS2 example. 59-60: update and restore the previous version, inspect the archive and review responsible engineering decisions together.

Hours 61-65: theory, 2+2+1

61-62: PLC, robot controller, industrial bus and cycle coordination; channel purposes. 63-64: official scopes of ISO 10218, ISO 13849 and IEC 61508, requirements and supporting results. 65: change management, engineering review and planning teamwork experience. Introductory hours cover terminology and the limits of standards' applicability; a full compliance analysis requires separate work.

Hours 66-75: practical session 4

66-67: Task 9, cell sketch, I/O and cycle coordination. 68-69: checks for cycle_id, missing parts, communication loss and late done. 70-71: Task 10, requirements and a change request for one parameter. 72-73: scenarios S1-S8, repeat the transfer after the change and restore the previous version. 74-75: final materials package, demonstration and handover to OPT120 and subsequent modules. The intervals total 75 hours.

How to reduce scope

If time is short, first remove the second object, complex contact model and interface decoration; then reduce the number of configuration variants. Keep a reproducible experiment with one manipulator, repeatability testing, operator intervention, the HRI study, threat and hazard assessment, industrial diagram and a requirement with its test. Move hand development, tactile sensors and force experiments to the separate OPT120 block, preserving M17 time for ethics and teamwork.

Acceptance checks and preparation for the hand project

Evidence for the four areas

Manipulation: scene, configuration, complete cycle or explicitly labelled kinematic demonstration, FK and IK verification, and a table of repeated trials. HRI: modes, cancellation, indications and a protocol for the participant study or self-conducted walkthrough. Ethics and cybersecurity: data inventory, use decisions, threat model and results of isolated tests. Industrial applications: cell diagram, I/O, requirements, change request and a plan to develop teamwork experience.

Prepare a README and a unified set of commands to reproduce the work. State in the report what was measured physically, what was tested in simulation and what was considered only theoretically. These experiments do not establish ISO/IEC certification, safe collaborative operation or industrial readiness. Save observed outcomes for the eight scenarios, and supporting results or a description of the remaining limitation for every significant protective measure.

Self-check questions

1. How does the tool pose differ from the object pose? 2. Why can IK have several valid solutions?
3. How does an attached object affect collision checking? 4. Why does attach not measure grasping force? 5. How does repeatability differ from accuracy? 6. How does the operator know who currently controls the robot?
7. What happens after cancel and communication loss? 8. How does a hazard differ from a cybersecurity threat? 9. Why does encryption not replace limit checks? 10. What does a hash prove about an update, and what does it not prove? 11. How does cycle_id prevent a false done? 12. What external experience do you still need to work in a robotics team?

What to hand over to OPT120

Hand over coordinate-frame descriptions, the baseline arm model, scene with objects, grasp parameters, reproducible action sequence, operator-intervention requirements and failure log. With this foundation, the separate 120-hour block after M17 develops hand mechanics, tactile perception, retention and physical testing. Its hours, equipment and criteria are accounted for separately; a completed anthropomorphic hand is not an M17 deliverable.

For the walking robot, retain the rules for mode selection, command authority, command-freshness checks and review of limits. They will be needed for external control of the body and legs. Before use on a real robot, test these decisions with the relevant mechanics, loads and actuator electronics. M17 is complete when you can explain and reproduce the entire testbed's operation, including failure and recovery.

Assigned reading: manipulation and human-robot interaction

Read the specified extracts before the corresponding tasks. Core reading explains principles; reference reading supports implementation. Sources may be in English; prepare your explanations and report in English. Reading is included in the 75 hours.

Core reading

1. Russ Tedrake, Robotic Manipulation (MIT, course notes). Basic Pick and Place: Monogram Notation; Pick and place via spatial transforms. For Tasks 1-2: write down the object → grasp → tool transforms and one pre-grasp pose; other chapters are not needed now.

2. Christoph Bartneck et al., Human-Robot Interaction: An Introduction, 2nd ed., 2024. Chapters Design, Research Methods, Robots in Society. For Tasks 4-6: one mode-indication principle, a pilot-study protocol and a data decision; skim Spatial Interaction to define the boundaries of the motion zone.

3. Ross Anderson, Security Engineering, 3rd ed. Ch. 4 Protocols; Ch. 6 Access Control. For Tasks 7-8: selected examples of authentication, message replay and permissions. After reading, distinguish an unauthorised command, an old command and an incorrect but permitted parameter.

Mechanics reference reading

4. Kevin Lynch, Frank Park, Modern Robotics. §12.1.2 Contact Types; §12.2.1 Friction; §11.5 Force Control. For Task 3 and the transition to OPT120: distinguish touching, slipping and holding, and position and force. Leave the full theory of force-based grasping to the separate block.

Laboratory documentation

Movelt Task Constructor: Pick and Place - stages and scene for Task 2; check main documentation against your pinned Jazzy version. SROS2: README and tutorials on the Jazzy branch - participant identification and permissions in Task 8, with permitted and prohibited communication.

Official ISO/IEC scope descriptions for Tasks 9-10 are listed on the preparation page. After reading, prepare a cell diagram, one verifiable requirement and applicability limits. Reading the standards in full is not required this month.

Fundamental concepts in M17

Pose. A body's position and orientation relative to a specified coordinate frame; both components are needed to describe a grasp.

Grasp frame. A coordinate frame on the object relative to which the desired end-effector position and orientation are specified.

Inverse kinematics (IK). Finding joint coordinates for a specified pose; a valid solution must respect limits and be checked using forward kinematics.

Pre-grasp pose (pregrasp). Position and orientation before approaching the object; it defines the approach direction and space for the open gripper.

Attached object. Geometry linked to the robot in the planning scene; this link alone does not establish physical retention of the object.

Repeatability. Closeness of repeated transfer results under identical conditions; a systematic offset may persist in every attempt.

Accuracy. Closeness of a result to a specified reference; it is evaluated relative to a defined goal, coordinate frame and measurement conditions.

Impedance. A specified dynamic relationship between motion and interaction force; it usually includes virtual stiffness and damping.

HRI. The study and design of human-robot interaction, including state understanding, control, errors and social context.

Mode error. An operator action based on an incorrect understanding of the current mode or of who controls the robot.

Hazard. A potential source of harm; the presence of a hazard alone does not describe an event's probability or the severity of its consequences.

Security threat. A possible violation of protected system properties, such as command substitution, data disclosure or loss of control availability.

Authentication. Verification of a participant's claimed identity or identifier by a trusted method; it does not itself define permitted actions.

Authorisation. A decision to permit a particular action by an identified participant, considering role, resource and access policy.

Encryption. Transforming data to conceal their content from those without the key; it does not replace checks of command authority, freshness and admissibility.

Hash. A short computed representation of content; comparison detects change relative to a trusted reference, but does not establish authorship.

Digital signature. Cryptographic verification of integrity and of a message's association with the signer's key, assuming trust in the corresponding public key.

Idempotency. A processing property whereby repeating the same logical command does not repeat the physical action or change.

PLC. A programmable logic controller that executes industrial cycle logic and processes inputs and outputs according to its configuration.

Cycle handshake. An exchange of readiness, request and completion states under defined rules; cycle_id links acknowledgement to a particular operation.

Equipment and software for M17

Use existing equipment

MacBook Pro M4; Ubuntu 24.04 ARM64 in UTM; ROS 2 Jazzy, Gazebo Harmonic and the MoveIt 2 configuration from M16. A virtual arm, table and object suffice for simulation. The M14 camera is needed only for the existing data in Task 6; do not repurchase sensors or the printer.

No mandatory purchases

M17 requires no purchases. Simulation is sufficient for all ten tasks; a physical arm, PLC, industrial bus and tactile sensors are unnecessary. The operator interface uses the keyboard and screen of the existing computer. A volunteer participant also needs no additional devices.

Install now: additions to the existing environment

In Ubuntu ARM64, add a Jazzy-compatible MoveIt Task Constructor only if it is absent and selected for Task 2; see [Pick and Place](#). Pin the package, commit and dependencies, and check main against Jazzy. A custom sequence of ROS actions is acceptable.

For Task 8, add [SROS2 tools and tutorials from the corresponding branch](#) in a separate ROS namespace. Save the chosen DDS implementation and its security settings. The mode FSM can be implemented in the existing code; a separate BehaviorTree library is needed only if a BT implementation is selected.

Installation checks

Launch the previous module's MoveIt scene, solve IK for one reachable pose, and execute a short transfer and cancel. In SROS2 with security enabled, check permitted communication and rejection of a participant without permission; keep keys out of the report. Test the graphical interface and networking inside UTM on your own computer, and save versions and logs.

Only after M17: the separate OPT120 block

Before building and testing the hand, prepare CAD, a BOM and requirements. Include one actuator and one selected contact or tactile sensor in the first order for a single-finger prototype, with suitable load capacity, power supply and interface. Determine quantities for the full hand after testing the finger. Purchases and the 120 hours are accounted for separately.

For introductory Tasks 9-10: [ISO 10218-1](#), [ISO 13849-1](#), [IEC 61508-1](#) - official scopes. This overview does not assign a PL/SIL or establish installation compliance.

Preparation and checks use time already allocated within the original 75 hours to review reading and setup. Delivery of possible OPT120 equipment is outside the learning workload. Check the availability of equipment and installed components.

Motion fundamentals: fingers and grasping

Required: touching is not yet retention

Kinematics establishes whether contact has been reached; statics establishes whether admissible forces can balance the object's weight and moments. First draw the object and finger separately, showing pairs of equal and opposite forces and the moment arm. For the finger, f_{ext} always denotes the force exerted by the object on the robot; the robot's force on the object points in the opposite direction.

A simple retention calculation

Consider a stationary object between two symmetric vertical fingers. Each contact has a normal force N and upward friction no greater than μN ; the weight mg acts downward. For a centred object without an external rotational moment, $2\mu N \geq mg$ is necessary. This checks that particular vertical equilibrium; it does not prove retention against every possible disturbance.

With $m=0.20$ kg, $g=9.81$ m/s² and $\mu=0.50$: $mg=1.962$ N and $N \geq 1.962$ N per finger. Choosing $N=2.5$ N gives a maximum total friction force of 2.5 N. If μ falls to 0.25, the limit becomes 1.25 N and can no longer support the weight. Equality at the boundary provides no margin; μ here is an illustrative model parameter.

For a massless vertical link with a normal-force moment arm of 0.04 m, the holding torque magnitude is $0.04 \cdot 1.962 \approx 0.0785$ N·m; friction in this geometry acts along the link. A real finger requires its Jacobian, mass and configuration: in statics, $\tau_{\text{act}} + \tau_g + J^T f_{\text{ext}} = 0$, where τ_g denotes joint torques due to gravity. Changing the external force direction also changes the sign of the compensating torque.

Why one motor does not determine two angles

For a finger with two independent joints, one taut, inextensible tendon can impose only $\Delta l = r_1 q_1 + r_2 q_2$ for constant moment arms. With $r_1 = r_2 = 0.01$ m and $\Delta l = 0.005$ m, $q_1 + q_2 = 0.5$ rad: the pairs (0.1;0.4) and (0.3;0.2) both satisfy the command. The actual pose depends on springs, contact and dynamics. This illustrates underactuation; when an object stops one link, the distribution of motion in the other changes.

Two exercises for Tasks 2-3

1. In the object-transfer report, save a force diagram and a table for three values of μ and two masses. With $N=2.5$ N, predict by hand whether the object will be held or will slip. Compare with contact simulation if available; with simplified attach, label the result a model calculation. Planner success does not establish this equilibrium.

2. Plot the line $q_1 + q_2 = 0.5$ and add the contact condition $q_1 = 0.2$ rad: this leaves $q_2 = 0.3$ rad. Save the plot and explain which physical condition determined the configuration. A full-size hand will need separate mechanics and sensors; a calculation without purchases suffices for now. Do not send torque commands to position-controlled servos.

Time allocation: 1 hour from 7-8 for statics; 1 hour from 17-18 to check the table as part of Task 3, and 1 hour from 19-20 to explain failure. These are within the original 75 hours. Building the hand and working with tactile sensors belong to the separate OPT120 stage after M17.

Assigned reading: Lynch/Park, Modern Robotics §5.2 and §12.2.1 - torque and friction; Tedrake, Fully-actuated vs Underactuated Systems: Definitions - independent coordinates and available control inputs.